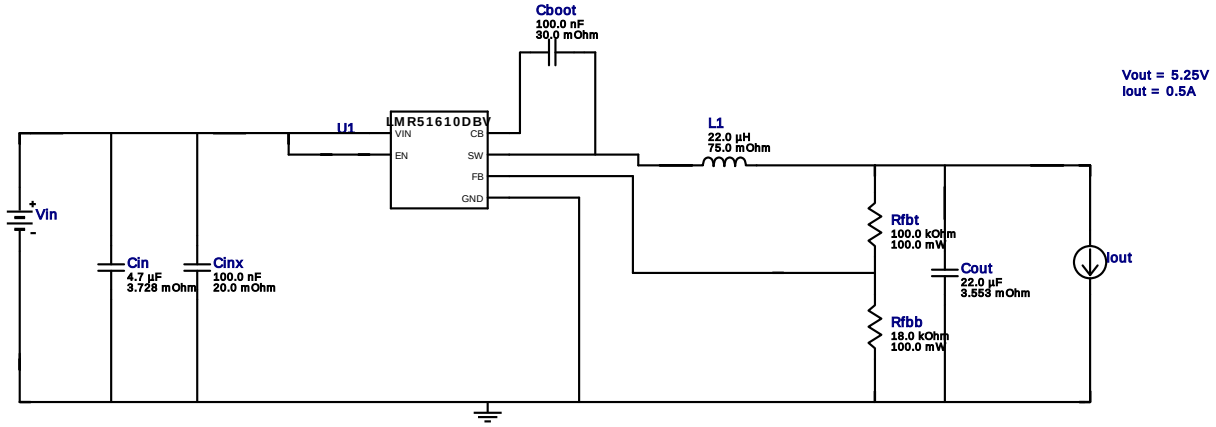
WEBENCH ® Design Report

VinMin = 9.0V
VinMax = 18.0V
Vout = 5.25V
Iout = 0.5A

Device = LMR51610DBVR
Topology = Buck
Created = 2025-07-25 22:48:28.773
BOM Cost = $0.99
BOM Count = 8
Total Pd = 0.22W

Design : 10 LMR51610DBVR
LMR51610DBVR 9V-18V to 5.33V @ 0.5A



Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM188R71E104KA01D Series= X7R	Cap= 100.0 nF ESR= 30.0 mOhm VDC= 25.0 V IRMS= 1.51 A	1	$0.01	0603 5 mm²
Cin	TDK	C1608X5R1V475K080AC Series= X5R	Cap= 4.7 uF ESR= 3.728 mOhm VDC= 35.0 V IRMS= 2.69359 A	1	$0.10	0603 5 mm²
Cinx	MuRata	GRM188R71H104KA93D Series= X7R	Cap= 100.0 nF ESR= 20.0 mOhm VDC= 50.0 V IRMS= 3.8 A	1	$0.02	0603 5 mm²
Cout	Taiyo Yuden	MSAST21GBB5226MTNC12 Series= X5R	Cap= 22.0 uF ESR= 3.553 mOhm VDC= 25.0 V IRMS= 2.17467 A	1	$0.22	0805 7 mm²
L1	TDK	VLP8040T-220M	L= 22.0 µH 75.0 mOhm	1	$0.22	VLP8040 113 mm²
Rfbb	Yageo	RC0603FR-0718KL Series= ?	Res= 18.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rfbt	Yageo	RC0603FR-07100KL Series= ?	Res= 100.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
U1	Texas Instruments	LMR51610DBVR	Switcher	1	$0.40	DBV0006A 15 mm²

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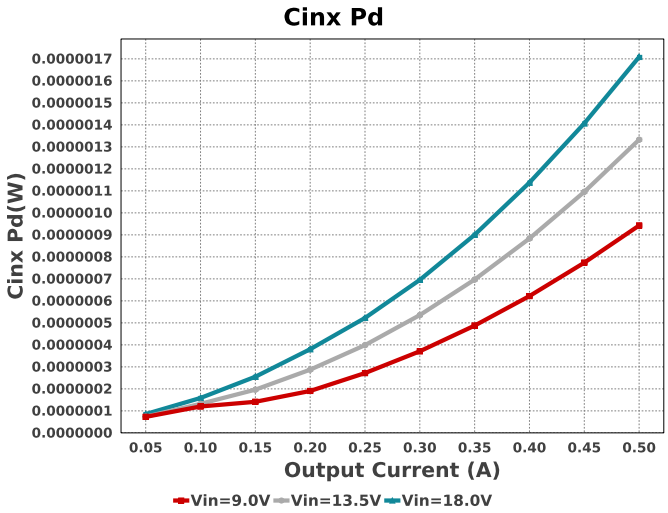

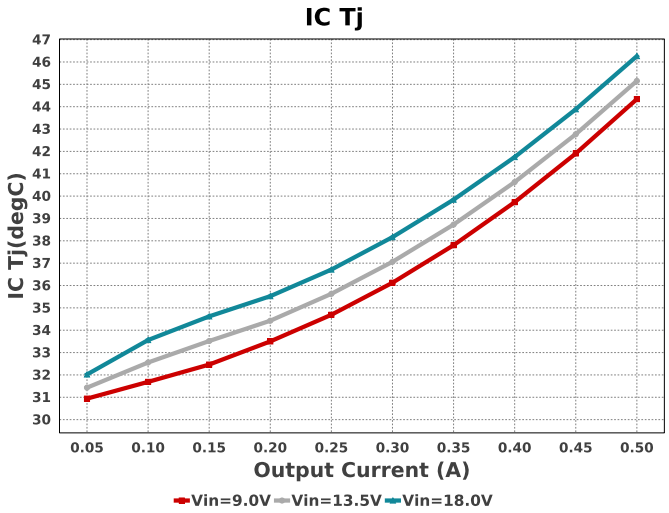

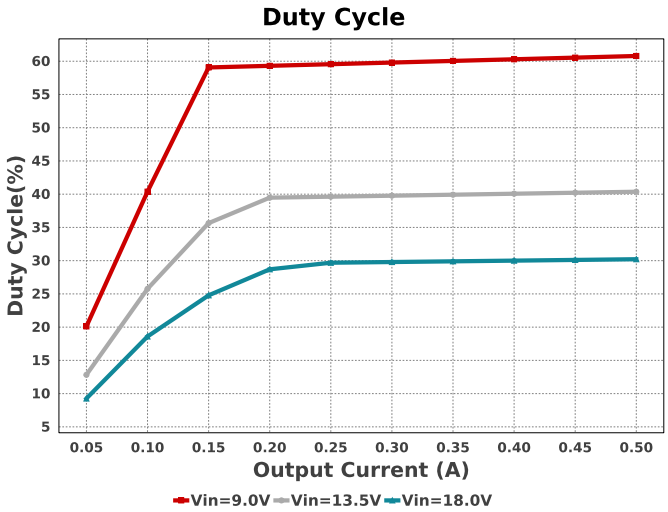

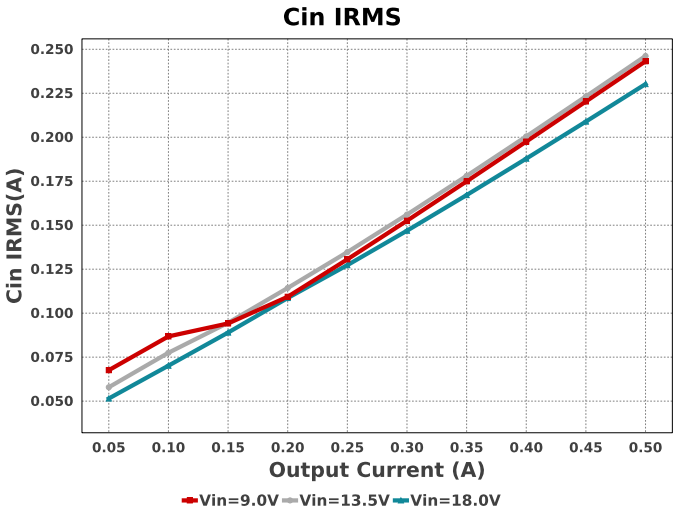

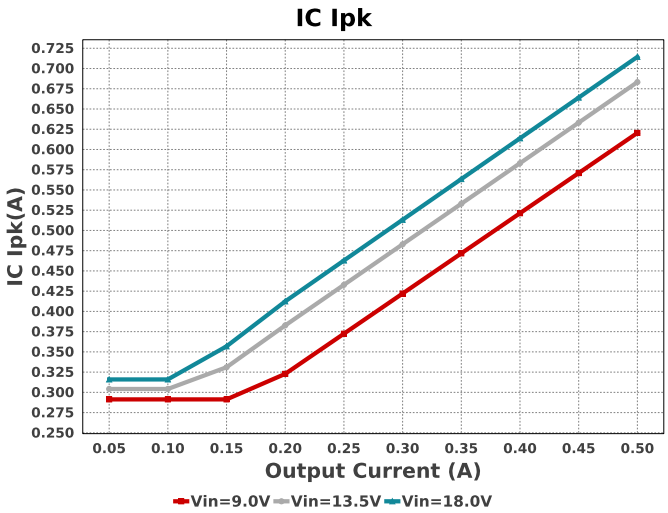

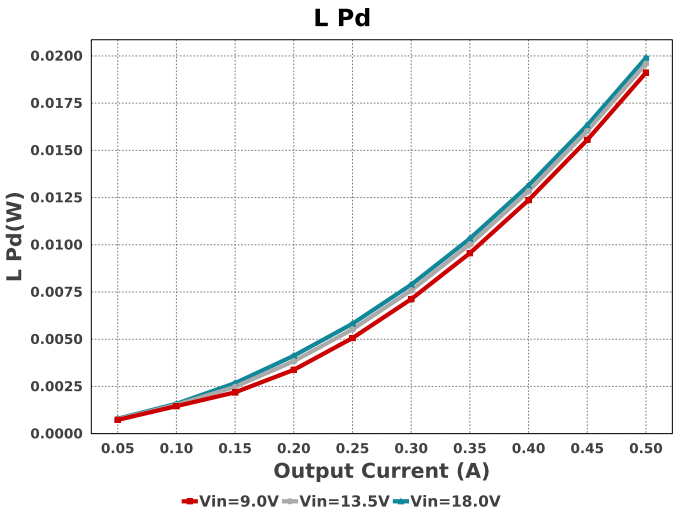

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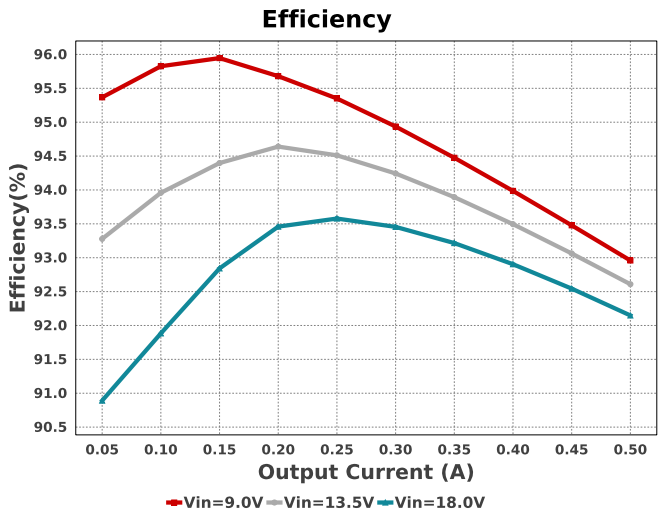
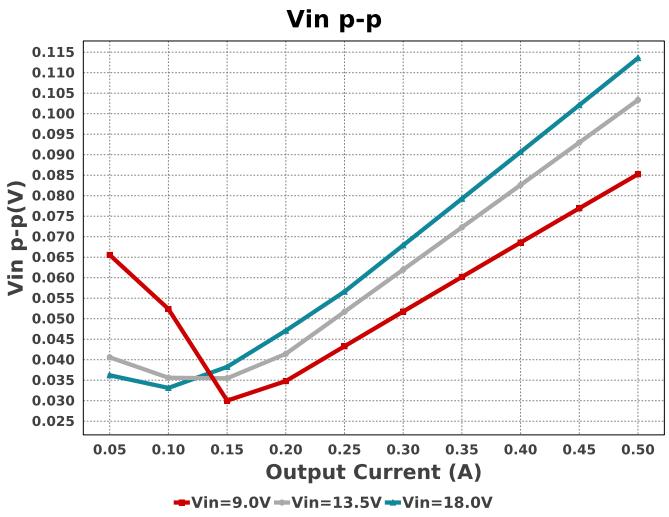
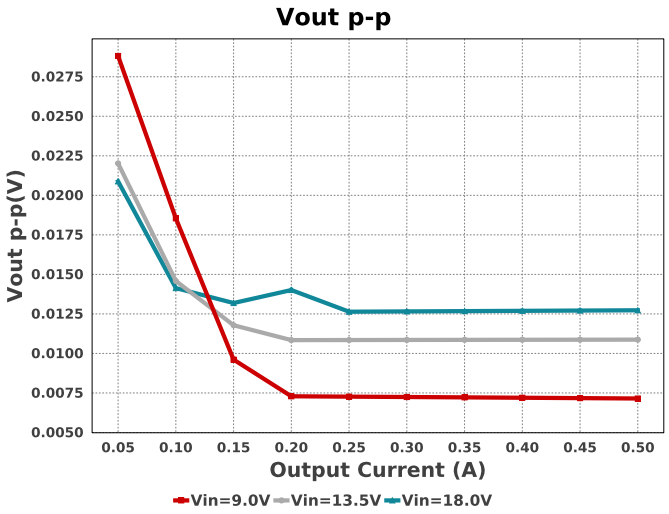
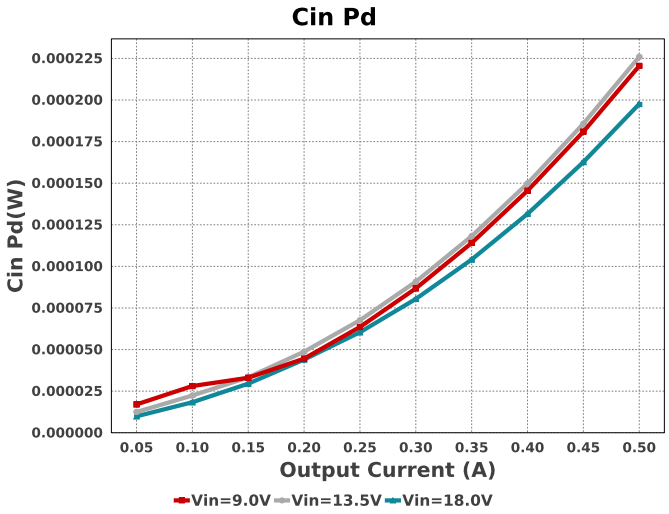
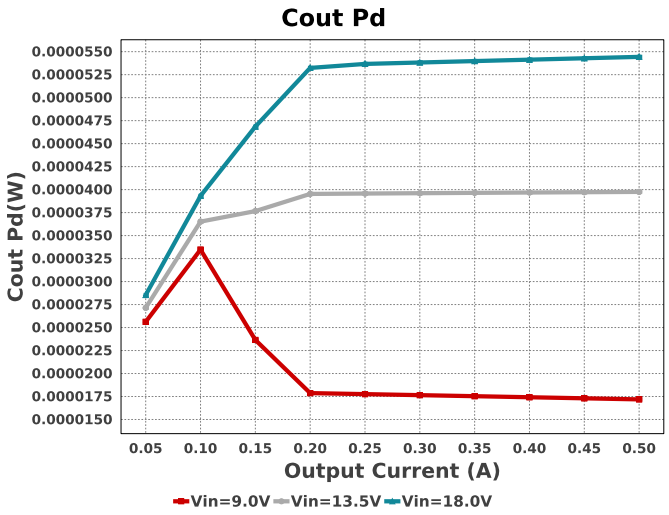
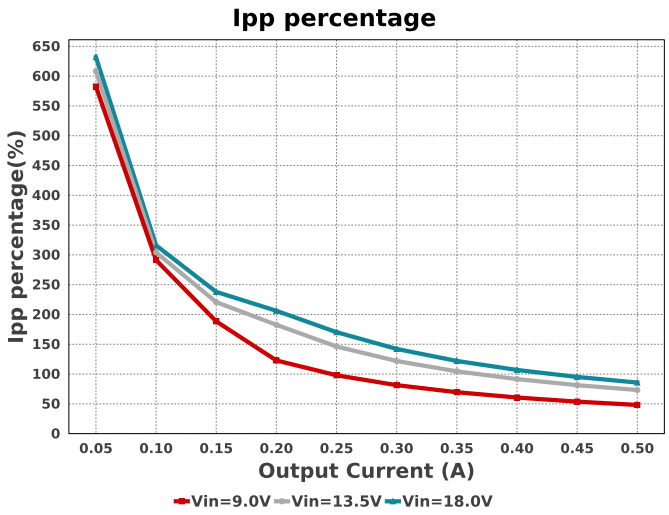

Efficiency
Efficiency(%)
96.0
95.5
95.0
94.5
94.0
93.5
93.0
92.5
92.0
91.5
91.0
90.5
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
Output Current (A)
Vin=9.0V Vin=13.5V Vin=18.0V

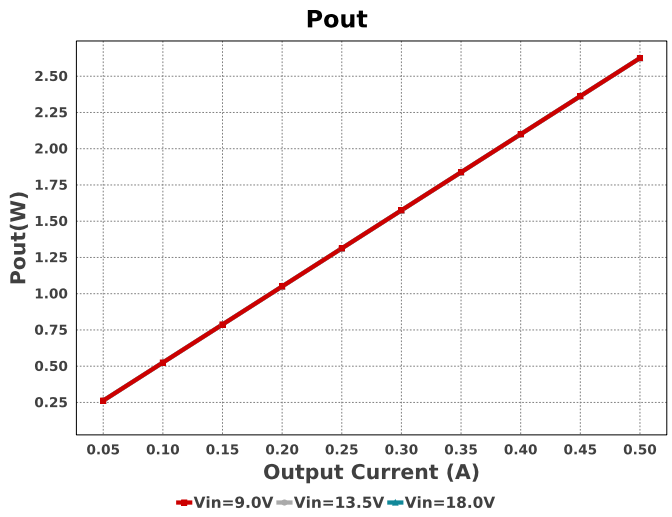

Vin p-p
Vin p-p(V)
0.115
0.110
0.105
0.100
0.095
0.090
0.085
0.080
0.075
0.070
0.065
0.060
0.055
0.050
0.045
0.040
0.035
0.030
0.025
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
Output Current (A)
Vin=9.0V Vin=13.5V Vin=18.0V

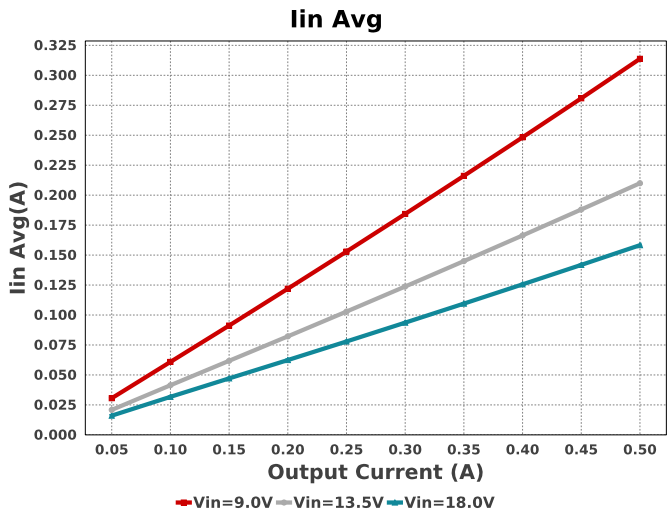

Vout p-p
Vout p-p(V)
0.0275
0.0250
0.0225
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
Output Current (A)
Vin=9.0V Vin=13.5V Vin=18.0V

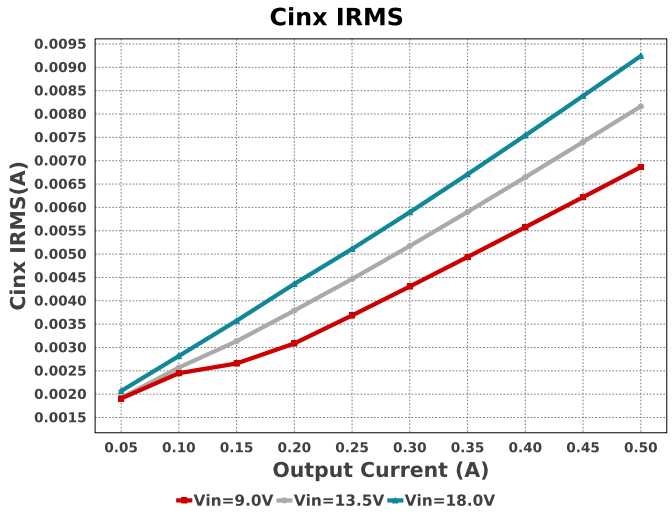

Cin Pd
Cin Pd(W)
0.000225
0.000200
0.000175
0.000150
0.000125
0.000100
0.000075
0.000050
0.000025
0.000000
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
Output Current (A)
Vin=9.0V Vin=13.5V Vin=18.0V

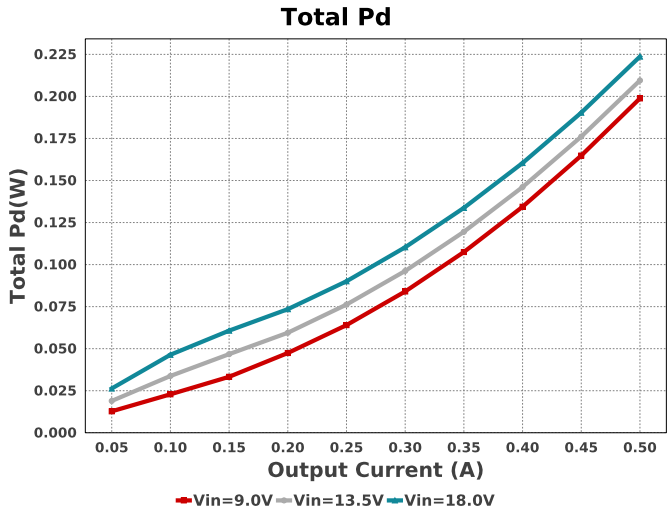

Cout Pd
Cout Pd(W)
0.0000550
0.0000525
0.0000500
0.0000475
0.0000450
0.0000425
0.0000400
0.0000375
0.0000350
0.0000325
0.0000300
0.0000275
0.0000250
0.0000225
0.0000200
0.0000175
0.0000150
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
Output Current (A)
Vin=9.0V Vin=13.5V Vin=18.0V

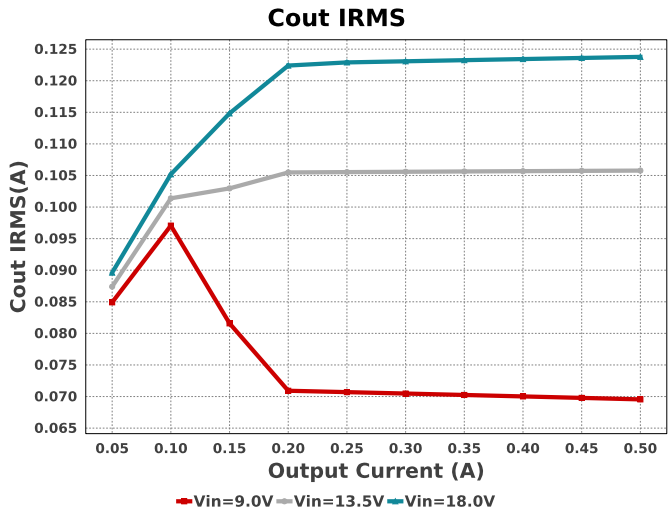

Ipp percentage
Ipp percentage(%)
650
600
550
500
450
400
350
300
250
200
150
100
50
0
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
Output Current (A)
Vin=9.0V Vin=13.5V Vin=18.0V

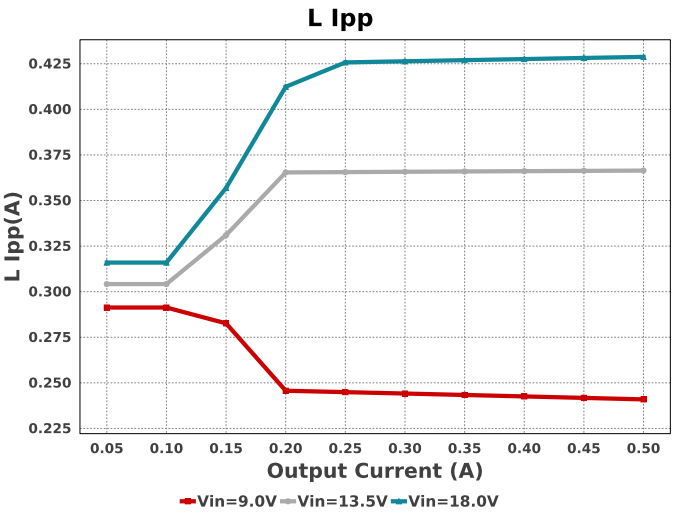












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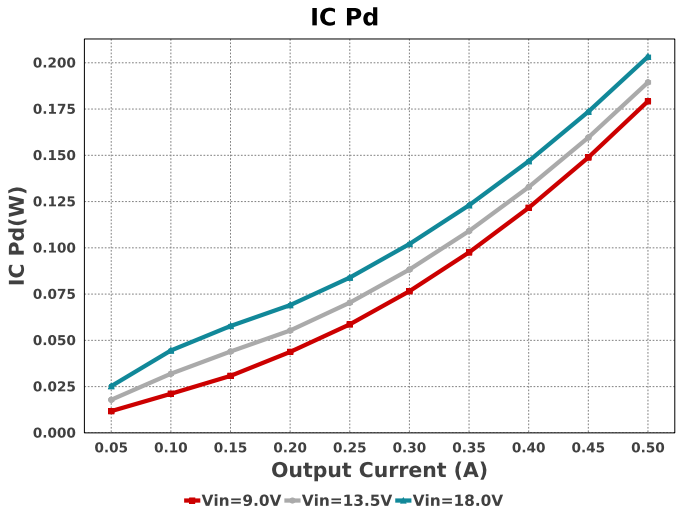

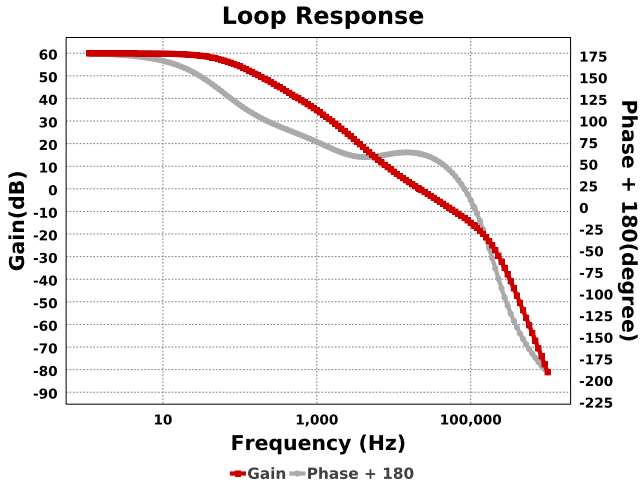

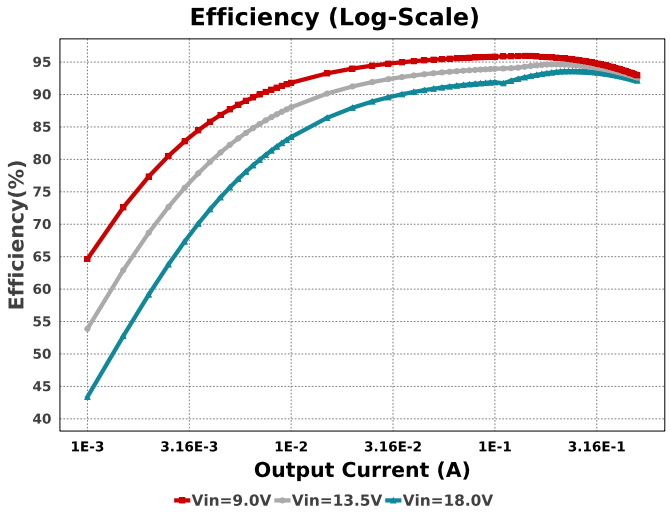

Operating Values

#	Name	Value	Category	Description
1.	BOM Count	8		Total Design BOM count
2.	Total BOM	$0.99		Total BOM Cost
3.	Cin IRMS	230.222 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	197.59 µW	Capacitor	Input capacitor power dissipation
5.	Cinx IRMS	9.243 mA	Capacitor	Bulk capacitor RMS ripple current
6.	Cinx Pd	1.709 µW	Capacitor	Bulk capacitor power dissipation
7.	Cout IRMS	123.777 mA	Capacitor	Output capacitor RMS ripple current
8.	Cout Pd	54.434 µW	Capacitor	Output capacitor power dissipation
9.	IC Ipk	714.388 mA	IC	Peak switch current in IC
10.	IC Pd	203.29 mW	IC	IC power dissipation
11.	IC Tj	46.263 degC	IC	IC junction temperature
12.	IC Tolerance	12.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA Effective	80.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
14.	Iin Avg	158.26 mA	IC	Average input current
15.	Ipp percentage	85.755 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
16.	L Ipp	428.776 mA	Inductor	Peak-to-peak inductor ripple current
17.	L Pd	19.899 mW	Inductor	Inductor power dissipation
18.	Cin Pd	197.59 µW	Power	Input capacitor power dissipation
19.	Cinx Pd	1.709 µW	Power	Bulk capacitor power dissipation
20.	Cout Pd	54.434 µW	Power	Output capacitor power dissipation
21.	IC Pd	203.29 mW	Power	IC power dissipation
22.	L Pd	19.899 mW	Power	Inductor power dissipation
23.	Total Pd	223.678 mW	Power	Total Power Dissipation
24.	Cross Freq	21.515 kHz	System Information	Bode plot crossover frequency
25.	Duty Cycle	30.215 %	System Information	Duty cycle
26.	Efficiency	92.148 %	System Information	Steady state efficiency
27.	FootPrint	158.0 mm^2	System Information	Total Foot Print Area of BOM components

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#	Name	Value	Category	Description
28.	Frequency	400.0 kHz	System Information	Switching frequency
29.	Gain Marg	-16.421 dB	System Information	Bode Plot Gain Margin
30.	Inductor ripple current requirement used for Inductor selection	40.0 %	System Information	Custom Inductor ripple current (% of average inductor current) requirement used for Inductor selection
31.	Iout	500.0 mA	System Information	Iout operating point
32.	Iout transient step used for Cout calculations	250.0 mA	System Information	Custom Transient current step requirement that was used for Cout selection (A).
33.	Low Freq Gain	59.897 dB	System Information	Gain at 1Hz
34.	Mode	CCM	System Information	Conduction Mode
35.	Overshoot Value	12.388 mV	System Information	Theoretical Vout Overshoot Value
36.	Phase Marg	63.132 deg	System Information	Bode Plot Phase Margin
37.	Pout	2.625 W	System Information	Total output power
38.	Undershoot Value	46.979 mV	System Information	Theoretical Vout Undershoot Value
39.	Vin	18.0 V	System Information	Vin operating point
40.	Vin p-p	113.561 mV	System Information	Peak-to-peak input voltage
41.	Vout	5.25 V	System Information	Operational Output Voltage
42.	Vout Actual	5.244 V	System Information	Vout Actual calculated based on selected voltage divider resistors
43.	Vout Ripple requirement used for Cout calculations	3.0 %	System Information	Custom maximum output ripple requirement that was used for Cout selection(% of Vout).
44.	Vout Tolerance	3.238 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
45.	Vout p-p	12.729 mV	System Information	Peak-to-peak output ripple voltage
46.	Vout transient requirement used for Cout calculations	3.0 %	System Information	Custom Transient voltage change requirement that was used for Cout selection (% of Vout).

Design Inputs

Name	Value	Description
Iout	500.0 m	Maximum Output Current
VinMax	18.0	Maximum input voltage
VinMin	9.0	Minimum input voltage
Vout	5.25	Output Voltage
base_pn	LMR51610	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

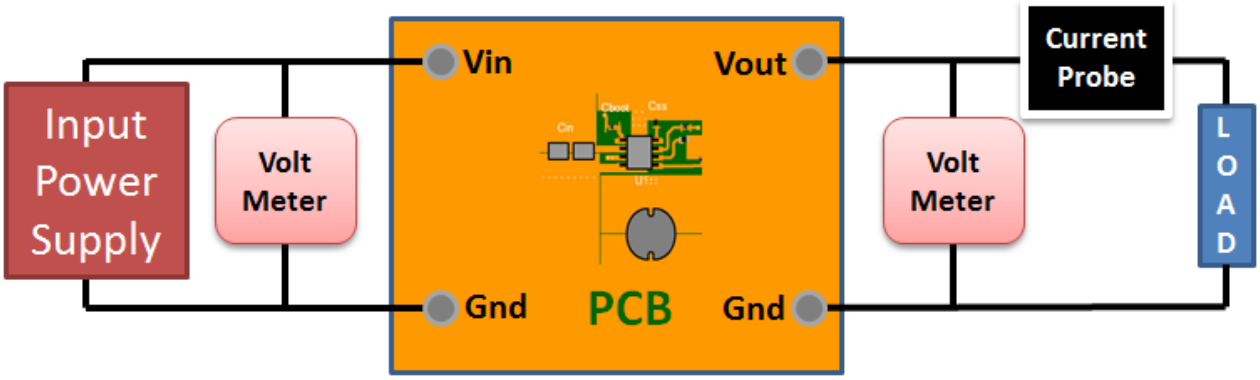
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 9.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 856C3ACFC3D186BD121B5CEC516967DA[v1]
2. **LMR51610** Product Folder : http://www.ti.com/product/LMR51610 : contains the data sheet and other resources.

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